Emanuel Medina Araujo

Software Engineer · Working Student at Vector · M.Sc. Informatics, TUM

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EXPERIENCE

Working Student Software Engineer — Vector Informatik GmbH
Nov 2024 – present

VSceneCreator, a 3D OpenDRIVE editor · Unity / C# · Munich

- Helped ship alpha versions **0.1** → **0.4** by designing and implementing features, fixing bugs and writing tests with Product Management.
- Architected a read-only **Explore Mode** with a hierarchical overview tree and dynamic tooltips, segregating editing features: a preview tool for licensed users and a better UX for free users.
- Restructured the base model to lazy-load non-essential and non-visible objects: **up to 70% faster** edit-mode switching.
- Optimized file loading through batched object creation and a progress bar replacing freezes: **10% faster** loading and lower memory use.

Working Student Software Engineer — Vector Informatik GmbH
Mar 2021 – Oct 2024

DYNA4, simulation environment for virtual driving tests · Java / Eclipse RCP · Munich

- Developed features, fixed critical bugs and enhanced the UI across **5 major releases** (5.0 → 9.0) with Product Management.
- Engineered a trace-signal alias view with a new file format, a custom persistence layer, and built-in validation with quickfixes.
- Expanded SWTBot/JUnit test suites and linked Jenkins CI results to JIRA tickets, reducing manual QA effort.

EDUCATION

M.Sc. Informatics — Technical University of Munich
Oct 2026 – present

B.Sc. Computer Science, application area Physics — Technical University of Munich
Oct 2020 – Aug 2026

Thesis: Learning-Based Inverse Kinematics. Completed alongside part-time engineering work at Vector.

Abitur — Herzog-Christoph-Gymnasium Beilstein
2012 – 2020

MINT-EC certificate with distinction

SELECTED PROJECTS

LoveNeverDies — Legal Loves Tech Hackathon, Bavarian Ministry of Justice challenge
Aug 2026

- Built an end-to-end encrypted estate assistant for bereaved families (React, TypeScript, Supabase) in ten days with a team of lawyers, guided by a **141-node** legal decision tree.
- Zero-trust design where the server only stores ciphertext: hybrid post-quantum key exchange and signatures (ML-KEM + X25519, ML-DSA + Ed25519), AES-256-GCM and a Shamir-secret-shared vault.
- Verified row-level security with **130 Vitest files** against real Postgres migrations (PGlite/WASM) and multi-user Playwright E2E tests.

SchafPlay — peer-to-peer multiplayer card game (PWA), live on GitHub Pages
2026

- Offline-first PWA for Schafkopf (React, TypeScript) with serverless multiplayer over WebRTC data channels via manual SDP exchange.
- Host-authoritative engine with per-player state redaction, heuristic AI opponents, and a seeded Playwright E2E suite in CI.

Learning-Based Inverse Kinematics — Bachelor thesis
Jun – Nov 2024

- Compared learned IK solvers (MLP, LSTM, hybrid and probabilistic heads, plus a PPO baseline) for 2- and 3-DOF manipulators in PyTorch, trained through differentiable forward kinematics. A direct-angle MLP solved 100% of 10,000 test problems within tolerance at about 155 µs per inference, outperforming every more complex variant.

6D Object Pose Estimation for Robotics — seminar paper (IEEE format)
Apr – Jul 2026

- Surveyed methods from geometric registration to zero-shot foundation models; identified hybrid architectures (zero-shot initialization plus fast local tracking) as the most viable direction.

SKILLS

Languages	C#, Java, TypeScript, JavaScript, Python, C/C++, SQL, MATLAB
Frameworks	Unity, React, React Native, PyTorch, Eclipse RCP, Supabase/PostgreSQL, WebRTC
Tooling	Git, Linux, GitHub Actions, Jenkins, Playwright, Vitest, JUnit, SWTBot, Rider, Jira

BEYOND CODE

Full Ironman finisher (3.8 km swim, 180 km bike, 42.2 km run) in 14:21:26 · Trombone and piano, advanced